98.8%
of broad gauge network electrified, but 75% electricity comes from thermal power

₹2 lakh cr.
Estimated savings by Indian Railways in last decade in fuel costs and logistics

₹20,000 cr.
in Budget 2025 for Small Modular Reactors, that could meet Railways' green fuel needs by 2033



ELECTRIFIED The Vande Bharat Express arrives at Srinagar station, Jan. '25

AFP

WHAT STARTED OUT A FEW YEARS AGO as a vague, lofty aspiration to be the world's first 'net-zero (on greenhouse gas emissions)' rail network by the end of this decade has now crystallised into a data-rich, executable plan for the Indian railways. The network finally stands just shy of 100 per cent electrification, and is already eyeing its next big leap: nuclear power.

Not in the locomotives, but in the energy mix that drives them. Indian Railways has, in recent months, initiated talks with the Department of Atomic Energy (DAE) along with the Nuclear Power Corporation of India Ltd (NPCIL) to gain access to nuclear energy—more specifically Small Modular Reactors (SMRs)—to provide a reliable portion of the 10 GW (Gigawatts) of power it will require each year by 2030. As Union railways minister Ashwini Vaishnaw says, "It's clean and can be cheaper than many other sources (of power), apart from being free from disruptions."

Grid to Grounded Power

Electric traction construction, which in 2014 was being executed at the rate of 1.4 km a day, has increased to 18 km a day. As of March 2025, the railways had electrified 68,701 km, or 98.8 per cent of its total 69,512 km broad-gauge network, which the department says has led to diesel consumption coming down by over 6,400 million litres annually. But there's an irony built into this: almost 75 per cent of the electricity fuelling these electric trains is still generated from thermal coal—the two worst possible words in sustainability conversations.

Which is why the Railways is now reconsidering where that power comes from. Renewable energy such as solar, wind and hydro power is being added at breakneck speed, as by 2029-30, the country will need 30,000 MWe (megawatts-electric) of renewable power capacity. However, renewables are intermittent. Trains do not have the option to stop and wait till the sun rises again or the wind picks up. Enter nuclear power.

SMRs are of particular interest to the Indian Railways due to their small footprint, safety attributes and ease of siting. Rather than outsized power stations traditionally built in India, SMRs offer quicker construction and an ability to be less invasive and more spread out—perfect for dedicated rail traction hubs.

Central to India's SMR programme is the 200 MWe Bharat Small Modular Reactor or BSMR-200, a pressurised heavy water reactor (PHWR) developed in India by a collaboration between the Bhabha Atomic Research Centre (BARC) and the NPCIL. Its pluses are that it is fuelled by slightly enriched Uranium (SEU), includes passive safety features and has provision for on-site spent fuel and waste storage. Also, the bulk of its critical equipment is already within the manufacturing capacity of Indian industry.

The lead BSMR unit will be established on a DAE site. The design has been submitted for approval, and should it come through and funds permit, the pilot project could be constructed in 60-72 months. In addition to BSMR-200, the DAE is also developing a twin SMR-55 plant with ultra-low land footprint and modular 55 MWe design,